

LakeTemperature Testing Software

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Background

- We are testing a module of *E3SM*,¹ an environmental model. This module, *LakeTemperature*, is part of the greater *ELM* (*E3SM* Land Model) which was originally a fork of the *CLM* (Community Land Model).
- We are able to test specifically this module using *SPEL*,² a software that adds a *NetCDF*³ interface for the constants, inputs, and outputs of a module of *E3SM* and compiles that specific part of the model. The executable produced by this is named `elmtest.exe`.

1. "Energy Exascale Earth System Model." *E3SM*, U.S. Department of Energy: Office of Science, n.d., e3sm.org/.

2. Schwartz, Peter. "SPEL_OpenACC." *GitHub*, 16 Oct. 2025, github.com/peterdschwartz/SPEL_OpenACC.git.

3. "NetCDF." *NSF UNIDATA*, n.d., www.unidata.ucar.edu/software/netcdf.

Project Goals

- Develop a software that can systematically change the constants used in the LakeTemperature model and allow us to visualize and observe patterns in how this affects its output.
- Apply testing techniques to try and find values for constants that will cause physically impossible or otherwise erroneous model outputs. Apply assertions pertaining to the values of model constants, inputs, and outputs based around physical laws and other constraints provided by domain scientists.

Our Testing Solution

- Automates the process of modifying model constants in `spell-constants0001.nc` and the process of analyzing the resulting values in `fut-outputs0001.nc`
- Both the process of choosing values for each model constant and the process of analyzing the resulting model outputs is extendable through sampling plugins and analysis plugins respectively.

Sampling Plugins

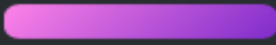
SampleGroup(s)

Testing

Tables of Testing Data

Analysis Plugins

Key

-  = Input Data
-  = Reference Data
-  = Output Data



Steps of Testing



Sample Generation



Using a Sampling Plugin to create samples
with unique values for the 25 scalar model constants



Running elmtest.exe



Running the program using the model constants
and recording the results



Analysis



Using an Analysis Plugin to analyze the results,
especially in comparison to reference data

Model Constants, Inputs, and Outputs

- *28 model constants* control how the model behaves and manipulates data.
 - We don't yet support modification of the 3 array constants. Of the 25 scalar constants we experiment with, 8 have no observable effect on the outputs.
- *25 model inputs* provide starting data for the model.
- *9 model outputs* record data that the model has generated.
- Additional 8 variables serve as both model inputs and outputs

Test Inputs: Model Constants (28)

Array Constants (3)	crop, iscft, percrop
Scalar Constants with Observable Outputs (17)	betavis, cnfac, cpliq, denh2o, dtime_mod, grav, hfus, lake_no_ed, lakepuddling, nlevgrnd, nlevlak, nlevsno, nlevsoi, thk_bedrock, tkwat, use_lch4, vkc
Scalar Constants without Observable Outputs (8)	cpice, denice, depthcrit, iulog, mixfact, pudz, tkair, tkice

Test Outputs: Model Inputs & Outputs (42)

Model Inputs Only (25)	col_es%t_grnd, col_pp%dz, col_pp%dz_lake, col_pp%z, col_pp%z_lake, col_pp%zi, col_pp&lakedepth, col_pp&snl, col_ws%h2osno, col_ws%h2osoi_ice, col_ws%h2osoi_liq, lakestate_vars%etal_col, lakestate_vars%ks_col, lakestate_vars%lake_raw_col, lakestate_vars%ws_col, soilstate_vars%csol_col, soilstate_vars%tkmg_col, soilstate_vars%tksat_col, soilstate_vars%watsat_col, solarabs_vars%fsds_nir_d_patch, solarabs_vars%fsds_nir_i_patch, solarabs_vars%fsr_nir_d_patch, solarabs_vars%fsr_nir_i_patch, solarabs_vars%sabg_lyr_patch, solarabs_vars%sabg_patch
Model Inputs/ Outputs (8)	col_es%t_lake, col_es%t_soisno, lakestate_vars%lake_icefrac_col, lakestate_vars%lake_icethick_col, veg_ef%eflx_gnet, veg_ef%eflx_sh_grnd, veg_ef%eflx_sh_tot, veg_ef%eflx_soil_grnd
Model Outputs Only (9)	ch4_vars%grnd_ch4_cond_col, col_ef%errsoi, col_es%hc_soi, col_es%hc_soisno, col_wf%qflx_snofrz, col_ws%frac_iceold, lakestate_vars%betaprime_col, lakestate_vars%lakeresist_col, lakestate_vars%savedtke1_col

NetCDF

- Network Common Data Form (*NetCDF*) is a format for self-describing datasets. Each file is composed of “dimensions” and “variables.” Variables are multi-dimensional arrays, with the meaning of each dimension described in the file.

```
netcdf x_times_y {  
  dimensions:  
    x = 5 ;  
    y = 5 ;  
  variables:  
    float x(x) ;  
    float y(y) ;  
    float z(x, y) ;  
  data:  
  
    x = -2, -1, 0, 1, 2 ;  
  
    y = -4, -2, 0, 2, 4 ;  
  
    z =  
      8, 4, -0, -4, -8,  
      4, 2, -0, -2, -4,  
      -0, -0, 0, 0, 0,  
      -4, -2, 0, 2, 4,  
      -8, -4, 0, 4, 8 ;  
}
```

What is a Sample?

- A sample is a set of values for the model constants contained in `spel-constants0001.nc`.
- `elmtest.exe` is run once for each sample.

What is a Sample? (Illustration)

Model Constants (formerly lakeparams.txt)

```
betavis
0.0
za_lake
0.6
n2min
7.4999999999999993E-00
5
tdmax
277.0
pudz
0.0
mixfact
10.0
depthcrit
25.0
```

Example Sample (JSON used for illustration)

```
{
  "betavis": 0.0,
  "za_lake": 0.6,
  "n2min": 7.4999999999999993E-005,
  "tdmax": 277.0,
  "pudz": 0.0,
  "mixfact": 10.0,
  "depthcrit": 25.0
}
```

What Outputs are Received per Sample?

- Analysis plugins will receive the model outputs from the `spe_l-outputs0001.nc` file in the form of multidimensional numpy arrays, which range from one to three dimensions.

Outputs Received by Analysis Plugins (*JSON* for illustration)

```
{  
  "veg_ef%eflx_soil_grnd": [0.0, ...],  
  "veg_ef%eflx_gnet": [[0.0, ...], ...],  
  ...  
}
```

Sample Groups

- Represent a grouping of ordered samples
 - When the program is run on each sample in a sample group, the model data is returned in the same order as the samples were in.
- ~~Analysis plugins can analyze the result of multiple consecutive runs of elmtest.exe.~~
- Having multiple sample groups is simply a convenient way of organizing similar samples.

Sample Group Illustration

Sample Group Data (JSON for illustration)

```
[
  {
    "betavis": 0.0,
    ...
  },
  {
    "betavis": 0.25,
    ...
  },
  {
    "betavis": 0.5,
    ...
  }
]
```

Ordered Output Data (JSON for illustration)

```
[
  {
    "lakestate_vars%betaprime_col": [0.0, ...],
    ...
  },
  {
    "lakestate_vars%betaprime_col": [0.25, ...],
    ...
  },
  {
    "lakestate_vars%betaprime_col": [0.5, ...],
    ...
  }
]
```

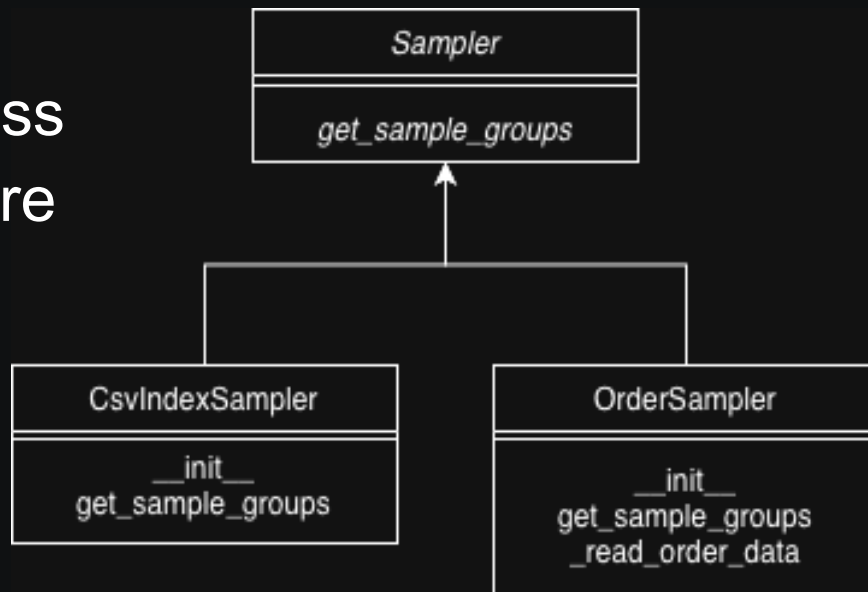
Plugins

- Our testing framework supports two types of plugins, sampling and analysis. Sampling plugins define values for model constants, and analysis plugins view the model outputs.
- Two sampling plugins are currently supported:
 - CSV Index Sampling, Order Sampling
- Four analysis plugins are currently supported:
 - Change Detection, Differences, Fault Finding, NetCDF4 Output

Sampling Plugin Overview

```
class Sampler(ABC):  
    @abstractmethod  
    def get_sample_groups(self) -> Mapping[str, SampleGroup]:  
        pass
```

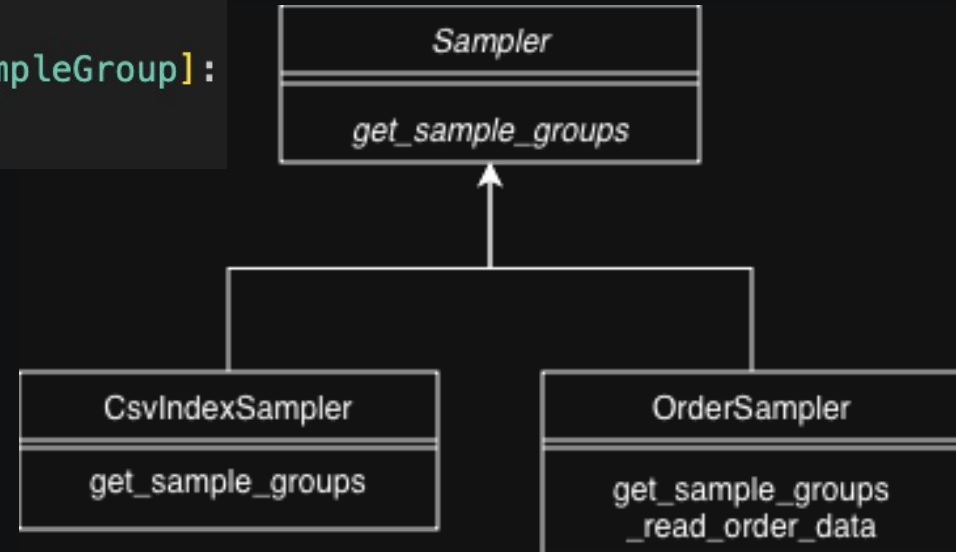
- Sampling plugins define a subclass of `Sampler`, returning one or more `SampleGroups`.



Sampling Plugin Overview

- Sampling plugins define a subclass of `Sampler` and return one or more `SampleGroups`.

```
class Sampler(ABC):  
    @abstractmethod  
    def get_sample_groups(self) -> Mapping[str, SampleGroup]:  
        pass
```



Our Sampling Plugins (2)

CSV Index Sampling	Creates samples using CSV files generated by <i>NIST's ACTS</i> ¹ tool by linearly interpolating between the values for each constant found in the ranges file
Order Sampling	Generates Order(s), a subclass of SampleGroup. Each Order reads instructions from a <i>JSON</i> file to generate a samples using linear interpolation. Each Order will be one sample group.

1. "Automated Combinatorial Testing for Software (ACTS)." *National Institute of Standards and Technology*, 20 July 2016, [csrc.nist.gov/SNS/acts/](https://csrc.nist.gov/groups/SNS/acts/).

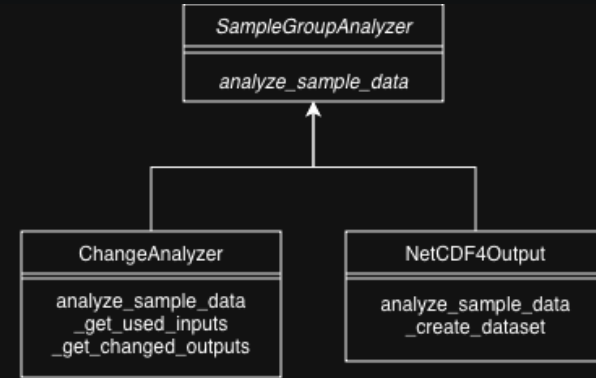
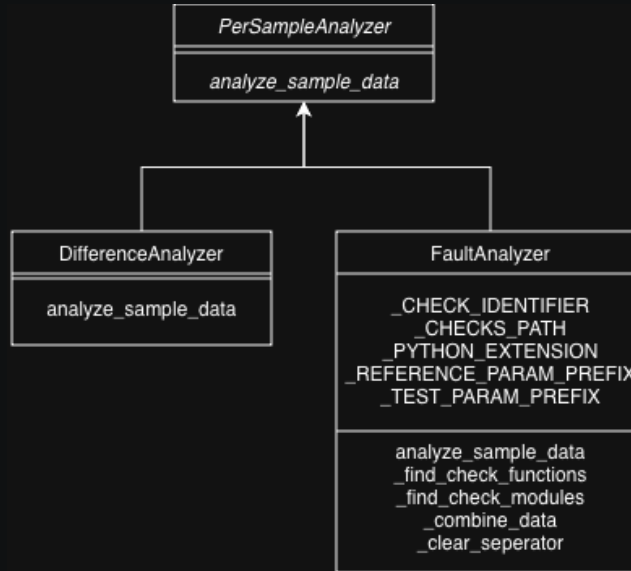
Analysis Plugin Overview

- Analysis plugins can either be “per sample” or “sample group” analysis plugins.
- “Per sample” plugins define a subclass of `PerSampleAnalyzer`, whose method is called each time `elmtest.exe` is run, and which receives the model outputs produced by that specific sample.
- “Sample group” plugins define a subclass of `SampleGroupAnalyzer`, whose method is called after all samples in a group are run through `elmtest.exe`, and which receives all model outputs from each.

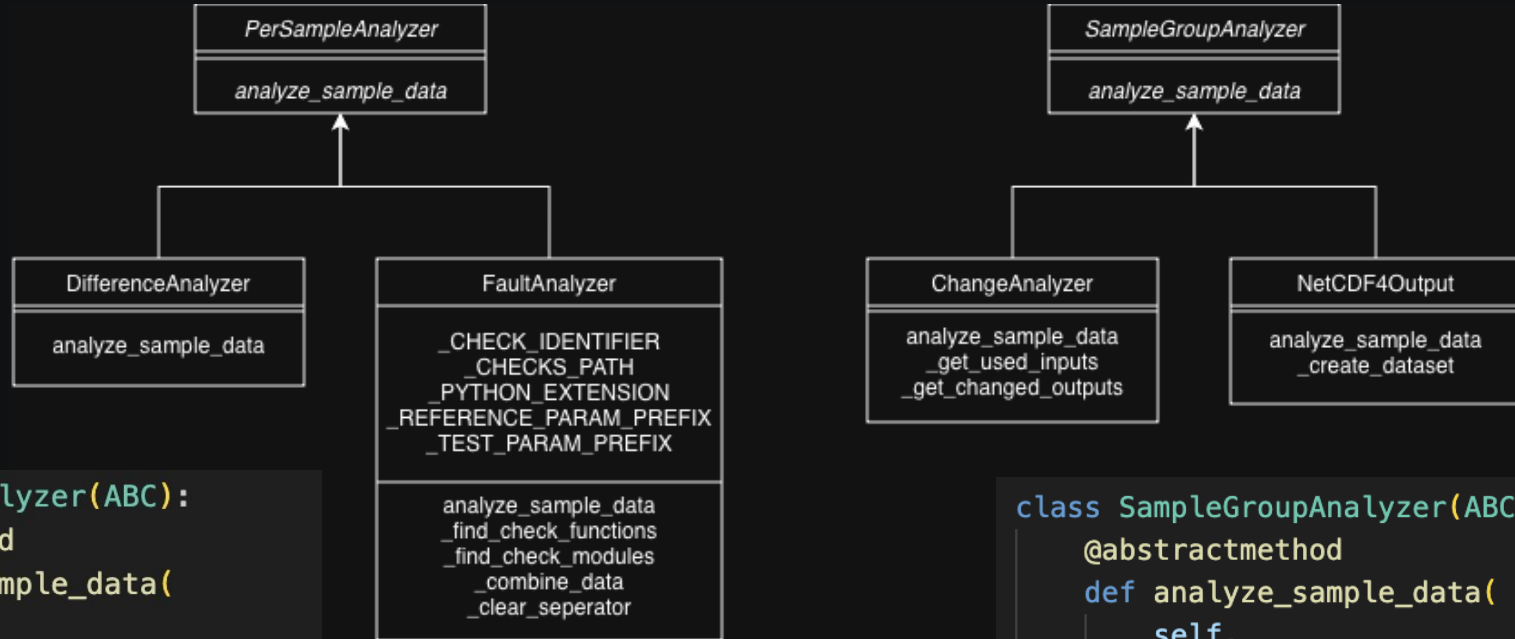
Analysis Plugin Overview (Cont.)

```
class PerSampleAnalyzer(ABC):  
    @abstractmethod  
    def analyze_sample_data(  
        self,  
        sample: Sample,  
        reference_data: ModelData,  
        test_data: ModelData,  
    ) -> tuple[str, FileSystemTree]:  
        pass
```

```
class SampleGroupAnalyzer(ABC):  
    @abstractmethod  
    def analyze_sample_data(  
        self,  
        sample_group: SampleGroup,  
        reference_data: ModelData,  
        sample_data: list[ModelData],  
    ) -> tuple[str, FileSystemTree]:  
        pass
```



Analysis Plugin Overview (Cont.)



```
class PerSampleAnalyzer(ABC):
    @abstractmethod
    def analyze_sample_data(
        self,
        sample: Sample,
        reference_data: ModelData,
        test_data: ModelData,
    ) -> tuple[str, FileSystemTree]:
        pass
```

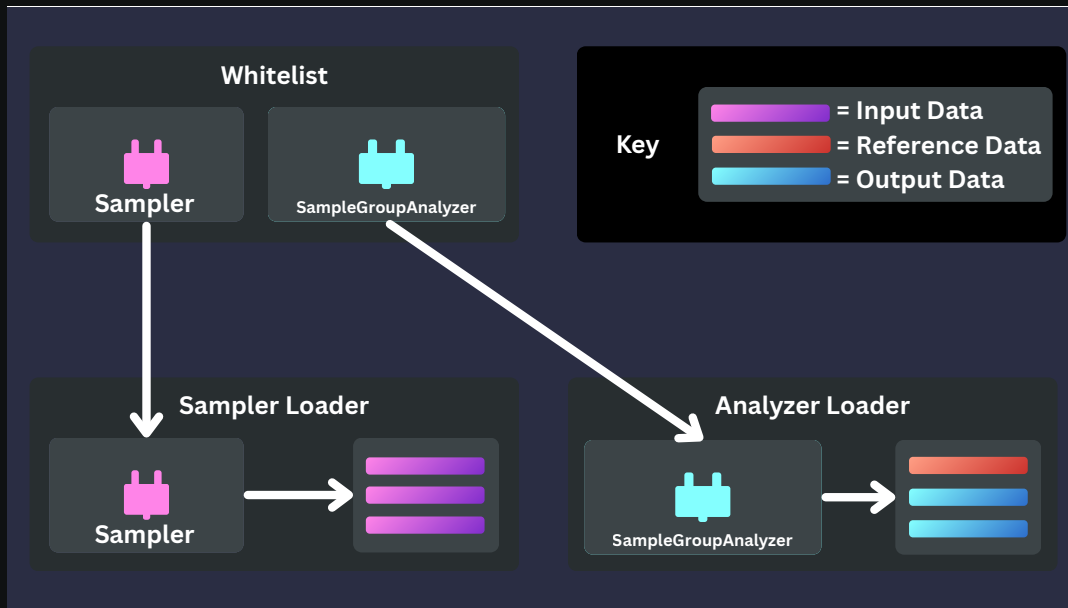
```
class SampleGroupAnalyzer(ABC):
    @abstractmethod
    def analyze_sample_data(
        self,
        sample_group: SampleGroup,
        reference_data: ModelData,
        sample_data: list[ModelData],
    ) -> tuple[str, FileSystemTree]:
        pass
```

Our Analysis Plugins (4)

Per Sample Analyzers	Differences	Finds differences between values in the reference data and the current sample's data, recording the indices of the difference, the two differing values, and the difference between them
	Fault Finding	Uses functions defined by the plugin to make assertions about values in the sample data, similar to a unit testing framework but with assertions pertaining to model values rather than specific function behavior
Sample Group Analyzers	Change Detection	Identifies which model inputs and outputs differ from the reference
	NetCDF4 Output	Stores each sample group as an individual <i>NetCDF</i> file, to be used for graphing changes based on our samples

Plugin Loader

- Is an abstract class that finds and loads whitelisted plugins



Example Whitelist

```
1  {
2      "sampling": [
3          "order_sampling"
4      ],
5      "analysis": [
6          "fault_finding"
7      ]
8  }
```


Sampler Loader

- defines a subclass of Plugin Loader
- Sampler Loader finds and loads the whitelisted Sampling Plugins.
- It then returns the Sample Groups they generate in a dictionary mapping the name of the Sample Group to itself.

```
class SamplerLoader(PluginLoader):
    def sample_from_plugins(self) -> dict[str, SampleGroup]:
        sample_groups: dict[str, SampleGroup] = {}
        samplers = self._plugin_objects[Sampler]

        for plugin_name in samplers:
            sampler = samplers[plugin_name]
            event_bus.fire_event(
                Event.BEGAN_SAMPLING_FROM_PLUGIN,
                plugin_name=plugin_name
            )

            try:
                sampler_sample_groups = sampler.get_sample_groups()
            except Exception as error:
                event_bus.fire_event(
                    Event.SAMPLING_FROM_PLUGIN_FAILURE,
                    plugin_name=plugin_name,
                    reason=error
                )
                continue

            self._add_new_sample_groups(
                sample_groups,
                sampler_sample_groups,
                plugin_name
            )

            group_sample_counts = {
                group_name: len(sampler_sample_groups[group_name])
                for group_name in sampler_sample_groups
            }

            event_bus.fire_event(
                Event.SAMPLING_FROM_PLUGIN_SUCCESS,
                plugin_name=plugin_name,
                group_sample_counts=group_sample_counts
            )

        return sample_groups
```

Analyzer Loader

- defines a subclass of Plugin Loader
- Analyzer Loader instantiates the Analysis Plugins and runs the analysis according to the plugin.

```
class AnalyzerLoader(PluginLoader):
    def run_sample_analysis(
        self,
        sample: Mapping[str, float],
        reference_data: Mapping[str, Sequence[float]],
        test_data: Mapping[str, Sequence[float]]
    ) -> None:
        for plugin_name, analyzer in self._plugin_objects[PerSampleAnalyzer].items():
            event_bus.fire_event(
                Event.BEGAN_SAMPLE_ANALYSIS_WITH_PLUGIN,
                plugin_name=plugin_name
            )

            try:
                console_out, file_out = analyzer.analyze_sample_data(
                    sample, reference_data, test_data
                )

                event_bus.fire_event(
                    Event.SAMPLE_ANALYSIS_WITH_PLUGIN_SUCCESS,
                    plugin_name=plugin_name,
                    console_output=console_out,
                    file_output=file_out
                )
            except Exception as error:
                event_bus.fire_event(
                    Event.SAMPLE_ANALYSIS_WITH_PLUGIN_FAILURE,
                    plugin_name=plugin_name,
                    reason=error
                )
```

```
class AnalyzerLoader(PluginLoader):
    def run_group_analysis(
        self,
        sample_group: SampleGroup,
        reference_data: ModelData,
        sample_data: list[ModelData],
    ) -> None:
        for plugin_name, analyzer in self._plugin_objects[SampleGroupAnalyzer].items():
            event_bus.fire_event(
                Event.BEGAN_GROUP_ANALYSIS_WITH_PLUGIN,
                plugin_name=plugin_name
            )

            try:
                console_out, file_out = analyzer.analyze_sample_data(
                    sample_group, reference_data, sample_data
                )

                event_bus.fire_event(
                    Event.GROUP_ANALYSIS_WITH_PLUGIN_SUCCESS,
                    plugin_name=plugin_name,
                    console_output=console_out,
                    file_output=file_out
                )
            except Exception as error:
                event_bus.fire_event(
                    Event.GROUP_ANALYSIS_WITH_PLUGIN_FAILURE,
                    plugin_name=plugin_name,
                    reason=error
                )
```

Configuration

- The LakeTemperature Testing Software and its plugins are controlled by configuration files.
- These files are loaded by different parts of the program and allow it to be adapted to different models and requirements.

Example Configurations

Main Configurations

```
[AppPaths]
app_root: .

[Model]
binary: model/elmtest
parameters: model/spel-constants0001.nc
parameter_defaults: model/spel-constant-defaults0001.nc
reference_output: model/spel-outputs0001.nc
test_output: model/fut-outputs0001.nc
args: 1
```

Plugin Whitelist Configurations

```
{
  "sampling": [
    "order_sampling"
  ],
  "analysis": [
    "fault_finding"
  ]
}
```

Output Configurations

Files

```
[Directories]
samples: testing_output/samples/
errors: testing_output/errors/
analysis: testing_output/analysis/
```

Logs

```
[Directories]
log_directory: logs/
```

```
[Settings]
keep_logs: 5
```

Sampling Plugin Configurations

CSV Index Sampling

```
[Paths]
range_path: model/FUT_lake_range.txt
order_directory: plugin_inputs/orders/
```

Order Sampling

```
[Paths]
range_path: model/FUT_lake_range.txt
csv_index_files_directory: plugin_inputs/csv_index_files/
```

Analysis Plugin Configurations

Fault Finding

```
[Paths]
checks_directory: plugin_inputs/fault_checks/
```

NetCDF4 Output

```
[Format]
store_as_samples: yes
```

Use Case Diagram

